

Multi-Gait Learning for Humanoid Robots Using Reinforcement Learning with Selective Adversarial Motion Prior

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Abstract—Learning diverse locomotion skills for humanoid robots in a unified reinforcement learning framework remains challenging due to the conflicting requirements of stability and dynamic expressiveness across different gaits. We present a multi-gait learning approach that enables a humanoid robot to master five distinct gaits—walking, goose-stepping, running, stair climbing, and jumping—using a consistent policy structure, action space, and reward formulation. The key contribution is a selective Adversarial Motion Prior (AMP) strategy: AMP is applied to periodic, stability-critical gaits (walking, goose-stepping, stair climbing) where it accelerates convergence and suppresses erratic behavior, while being deliberately omitted for highly dynamic gaits (running, jumping) where its regularization would over-constrain the motion. Policies are trained via PPO with domain randomization in simulation and deployed on a physical 12-DOF humanoid robot through zero-shot sim-to-real transfer. Quantitative comparisons demonstrate that selective AMP outperforms a uniform AMP policy across all five gaits, achieving faster convergence, lower tracking error, and higher success rates on stability-focused gaits without sacrificing the agility required for dynamic ones.

I. INTRODUCTION

Humanoid robots hold the promise of operating in environments designed for humans, but realizing this potential requires the ability to execute multiple locomotion behaviors adaptively [1]. Practical deployment demands not only walking for efficient traversal but also running for rapid movement, jumping for gap crossing or obstacle negotiation, stair climbing for multi-level navigation, and specialized gaits such as goose-stepping for structured or ceremonial scenarios. Each of these gaits imposes fundamentally different requirements on the control system: walking emphasizes energy efficiency and steady balance, running requires rapid force generation and aerial phase management, jumping demands maximal vertical impulse in a short time window, stair climbing needs precise foot placement and clearance control, and goose-stepping requires sustained leg extension with strict posture constraints. The central challenge is not merely acquiring each gait in isolation but learning them within a unified framework that shares policy architecture, observation space, and reward structure.

Reinforcement learning has emerged as a compelling approach for learning complex motor skills without requiring explicit analytical models. By interacting with a physics

simulator at scale, RL agents can discover control strategies that are difficult to hand-design, especially for underactuated, high-degree-of-freedom systems like humanoid robots [2]. The key advantage of RL over traditional model-based control is its ability to handle nonlinear dynamics, contact switching, and actuator limits in a unified optimization process. Recent work has demonstrated impressive single-gait capabilities, including robust walking on flat and uneven ground, stair traversal with foot clearance, and basic disturbance recovery [3], [4]. However, these studies typically train a separate policy for each task, and each policy is tuned with its own reward design and hyperparameter configuration. Extending RL to multiple gaits under a single framework introduces new difficulties: the policy must reconcile conflicting dynamics across behaviors, the reward function must remain effective across gaits with vastly different motion characteristics, and the observation space must encode sufficient information for all behaviors without becoming redundant or misleading for any individual gait [5].

Adversarial Motion Prior (AMP) addresses a key difficulty in RL-based locomotion—producing natural, stable motions that transfer well to hardware [6]. By training a discriminator to distinguish between agent-generated and reference motion clips, AMP regularizes the policy toward demonstrated human-like behavior distributions. This has been shown to reduce joint jitter, improve foot placement consistency, and produce smoother trajectories that are more compatible with real actuator dynamics [7], [8]. However, existing work that employs AMP does so uniformly: once an AMP module is introduced, it remains active for all behaviors throughout training. This raises an important but underexplored question: *is AMP always beneficial, or can it be counterproductive for certain gait types?*

Our observation is that the effectiveness of AMP depends critically on the nature of the target gait. AMP excels at reinforcing periodic, repetitive motion patterns with moderate joint amplitude—exactly the characteristics of walking, goose-stepping, and stair climbing. For these gaits, the reference motion distribution closely matches the desired policy output, and the adversarial reward provides a stabilizing inductive bias that accelerates convergence and suppresses erratic behavior. Conversely, for gaits requiring large joint excursions, rapid acceleration, and explosive dynamics—such as running and jumping—the available human motion references are limited, and the AMP discriminator’s regularization can act as an undesired constraint. In these cases,

基于选择性对抗运动先验的强化学习实现人形机器人多步态学习

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摘要 — 在统一的强化学习框架中为人形机器人学习多样化的运动技能仍然具有挑战性，因为不同步态在稳定性和动态表现力方面存在冲突需求。我们提出了一种多步态学习方法，使机器人能够使用一致的策略结构、动作空间和奖励公式掌握五种不同的步态——行走、正步、跑步、爬楼梯和跳跃。关键贡献在于一种选择性对抗运动先验（AMP）策略：AMP 应用于周期性、稳定性关键的步态（行走、正步、爬楼梯），在这些步态中它加速收敛并抑制异常行为，而在高度动态的步态（跑步、跳跃）中则有意省略，因为其正则化会过度约束运动。策略通过 PPO 在仿真中结合域随机化进行训练，并通过零样本仿真到现实迁移部署在物理 12 自由度人形机器人上。定量比较表明，选择性 AMP 在所有五种步态上均优于统一 AMP 策略，在稳定性重点步态上实现了更快的收敛、更低的跟踪误差和更高的成功率，同时不牺牲动态步态所需的敏捷性。

I. 引言

人形机器人有望在人类设计的环境中运行，但实现这一潜力需要具备自适应执行多种运动行为的能力 [1]。实际部署不仅需要行走以实现高效移动，还需要跑步以快速移动、跳跃以跨越间隙或越过障碍、爬楼梯以进行多层导航，以及诸如正步等用于结构化或仪式场景的专门步态。每种步态对控制系统提出了根本不同的要求：行走强调能效和稳定平衡，跑步需要快速发力并管理腾空阶段，跳跃要求在短时间内产生最大垂直冲量，爬楼梯需要精确的落脚点和间隙控制，而正步则需要持续的腿部伸展和严格的姿态约束。核心挑战不仅在于孤立地学习每种步态，更在于一个共享策略架构、观测空间和奖励结构的统一框架内学习它们。

强化学习已成为一种引人注目的方法，用于学习复杂的运动技能，而无需显式的分析模型。通过与物理

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模拟器进行大规模交互，强化学习智能体能够发现难以手工设计的控制策略，尤其适用于像人形机器人这样的欠驱动、高自由度系统 [2]。与传统基于模型的控制相比，强化学习的关键优势在于其能够在统一的优化过程中处理非线性动力学、接触切换和执行器限制。近期研究已展示了令人印象深刻的单一步态能力，包括在平坦和不平坦地面上的稳健行走、具有足部间隙的楼梯穿越以及基本的扰动恢复 [3], [4]。然而，这些研究通常为每个任务训练一个独立的策略，且每个策略都通过其自身的奖励设计和超参数配置进行调优。将强化学习扩展到单一框架下的多种步态引入了新的困难：策略必须协调不同行为之间的冲突动力学，奖励函数必须在运动特性差异巨大的步态间保持有效，且观测空间必须为所有行为编码足够的信息，而不会对任何单个步态变得冗余或具有误导性 [5]。

对抗性运动先验（AMP）解决了基于强化学习的运动控制中的一个关键难题——生成自然、稳定的运动，使其能够良好地迁移到硬件上 [6]。通过训练一个判别器来区分智能体生成的运动片段与参考运动片段，AMP 将策略正则化到所展示的类人行为分布上。这已被证明能减少关节抖动、改善足部放置的一致性，并产生与真实执行器动力学更兼容的平滑轨迹 [7], [8]。然而，现有采用 AMP 的工作是统一进行的：一旦引入 AMP 模块，它在整个训练过程中对所有行为都保持激活。这引出了一个重要但尚未充分探索的问题：AMP 是否总是有益的，或者对于某些步态类型它可能适得其反？

我们的观察是，AMP 的有效性关键取决于目标步态的特性。AMP 擅长强化具有中等关节幅度的周期性、重复性运动模式——这正是行走、正步走和爬楼梯的特征。对于这些步态，参考运动分布与期望的策略输出高度吻合，对抗性奖励提供了稳定的归纳偏置，加速收敛并抑制异常行为。相反，对于需要大关节活动范围、快速加速和爆发性动力学的步态（如奔跑和跳跃），可用的人类运动参考有限，AMP 判别器的正则化可能成为不必要的约束。在这些情况下，

the policy needs to explore beyond the reference manifold to discover strategies that fully utilize the robot’s actuator capacity, and the AMP penalty can prevent the agent from reaching the necessary motion extremes.

This mismatch motivates a selective approach: rather than applying AMP universally or discarding it entirely, we propose to use it where it helps and disable it where it hinders. This strategy requires understanding which gait characteristics align with the AMP assumption and which deviate from it, and making an informed decision for each locomotion mode.

The contributions of this paper are as follows:

- 1) **A unified RL framework** that learns five practical gaits for humanoid robots—walking, goose-stepping, running, jumping, and stair climbing—using a consistent policy architecture and reward function based on sinusoidal joint reference tracking. All gaits share the same observation space, network structure, and reward formulation, differing only in reference trajectory parameters and reward weights.
- 2) **A selective Adversarial Motion Prior strategy** that applies AMP to periodic, stability-critical gaits while omitting it for highly dynamic, agile gaits. We provide empirical analysis demonstrating that AMP improves convergence speed and tracking accuracy for walking-type gaits but restricts motion amplitude and expressiveness for running and jumping.
- 3) **Full real-robot validation** of all five gaits with quantitative comparisons including tracking error, convergence speed, success/fall rate, and AMP versus non-AMP performance. We deploy every learned policy on a physical humanoid platform, providing a reproducible real-robot baseline for multi-gait learning.

II. METHODOLOGY

A. Overview

We propose a unified reinforcement learning framework that learns five distinct locomotion gaits for a 12-DOF humanoid robot: walking, goose-stepping, running, jumping, and stair climbing. The framework consists of three main components. First, a unified RL formulation with a consistent observation space, action space, and reward structure is used to train all five gaits, where each gait is distinguished only by its sinusoidal reference trajectory parameters and reward weight configuration. Second, a selective AMP strategy is applied: AMP is enabled for periodic, stability-critical gaits (walking, goose-stepping, stair climbing) and disabled for highly dynamic gaits (running, jumping). Third, policies trained in simulation with extensive domain randomization are deployed directly on a physical humanoid robot without fine-tuning. Figure 1 illustrates the overall pipeline.

B. Reinforcement Learning Formulation

We formulate each gait learning task as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, r, \gamma, \rho_0)$. All five gaits share the same MDP structure, differing only in the

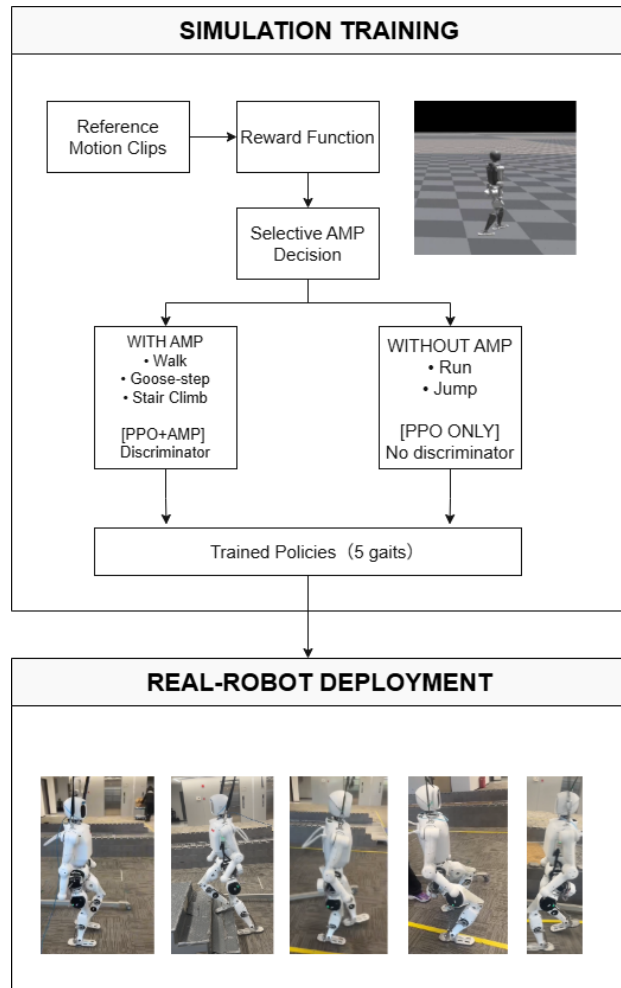


Fig. 1. Overview of the multi-gait learning pipeline. Policies are trained in simulation with or without AMP depending on gait type, then deployed on the physical robot.

reference trajectory generation, reward weight parameters, and AMP configuration.

1) *State Space*: The observation space received by the policy at each timestep t consists of proprioceptive states, reference signals, and command inputs. The per-frame observation vector $o_t \in \mathbb{R}^{50}$ is constructed as the concatenation of the following components:

- **Base linear velocity** $v_b \in \mathbb{R}^3$: expressed in the robot’s body frame, scaled by a factor of 2.0.
- **Velocity commands** $c \in \mathbb{R}^3$: target linear velocity in x and y directions and angular velocity about the z axis.
- **Phase encoding** $\phi \in \mathbb{R}^2$: sinusoidal encoding $[\sin(2\pi\phi_t), \cos(2\pi\phi_t)]$ of the current gait phase.
- **Base angular velocity** $\omega_b \in \mathbb{R}^3$: expressed in the body frame, scaled by 0.25.
- **Projected gravity vector** $g_p \in \mathbb{R}^3$: the world-frame gravity vector $[0, 0, -1]^\top$ rotated by the base orientation.
- **Joint position error** $q_e \in \mathbb{R}^{12}$: the difference between current joint positions and the default standing configuration.

策略需要探索参考流形之外的空间，以发现充分利用机器人执行器能力的方案，而 AMP 惩罚项可能阻止智能体达到必要的运动极限。

这种不匹配促使我们采用一种选择性方法：与其普遍应用 AMP 或完全舍弃它，我们建议在有益处的地方使用它，在有害处的地方禁用它。这一策略需要理解哪些步态特征符合 AMP 假设，哪些偏离了它，并为每种运动模式做出明智的决策。

本文的贡献如下：

- 1) **一个统一的强化学习框架**，该框架基于正弦关节参考轨迹跟踪，使用一致的策略架构和奖励函数，学习人形机器人的五种实用步态——行走、正步、跑步、跳跃和爬楼梯。所有步态共享相同的观测空间、网络结构和奖励公式，仅在参考轨迹参数和奖励权重上有所不同。
- 2) **一种选择性对抗性运动先验策略**，该策略将 AMP 应用于周期性、稳定性关键的步态，同时在高动态、敏捷的步态中省略它。我们提供的实证分析表明，AMP 能提高行走类步态的收敛速度和跟踪精度，但会限制跑步和跳跃的运动幅度与表现力。
- 3) **完整的真实机器人验证**，涵盖所有五种步态，并包括跟踪误差、收敛速度、成功 / 摔倒率以及 AMP 与非 AMP 性能的定量比较。我们在物理人形机器人平台上部署了每个学习到的策略，为多步态学习提供了可复现的真实机器人基线。

II. 方法论

A. 概述

我们提出一个统一的强化学习框架，用于训练一个 12 自由度人形机器人的五种不同运动步态：行走、正步走、跑步、跳跃和爬楼梯。该框架包含三个主要组成部分。首先，采用统一的强化学习公式，包含一致的观测空间、动作空间和奖励结构，用于训练所有五种步态，每种步态仅通过其正弦参考轨迹参数和奖励权重配置加以区分。其次，应用选择性对抗运动先验策略：对于周期性、稳定性关键的步态（行走、正步走、爬楼梯）启用对抗运动先验，而对于高度动态的步态（跑步、跳跃）则禁用。第三，在仿真中通过广泛域随机化训练的策略直接部署在物理人形机器人上，无需微调。图 1 展示了整体流程。

B. 强化学习公式

我们将每个步态学习任务建模为一个马尔可夫决策过程，由元组 $(\mathcal{S}, \mathcal{A}, r, \gamma, \rho_0)$ 定义。所有五种步态共享相同的马尔可夫决策过程结构，仅在

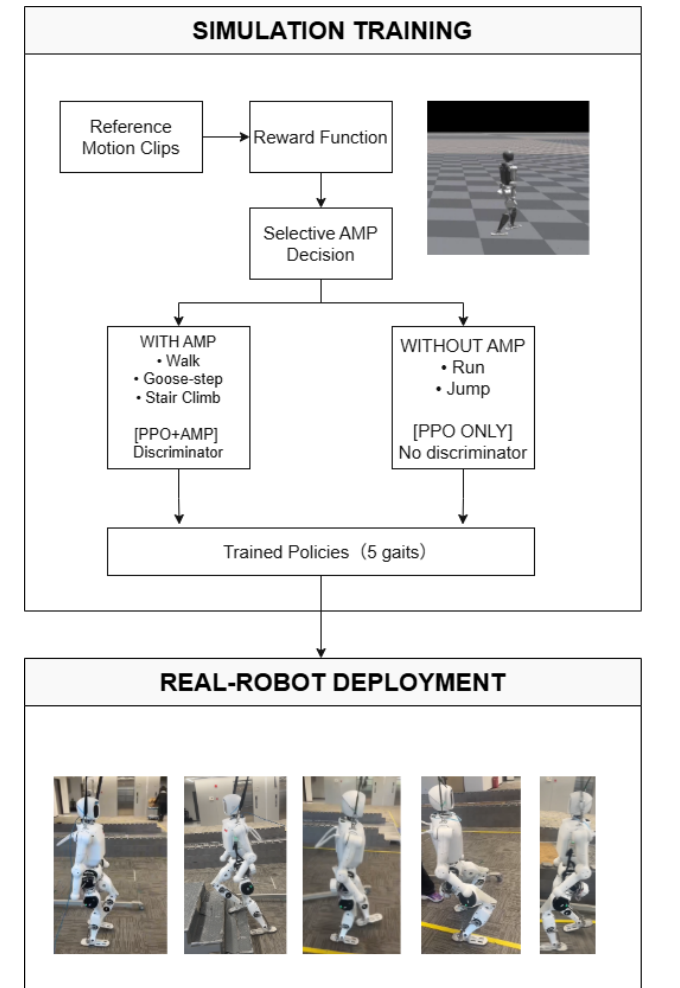


图 1. 多步态学习流程概览。根据步态类型，在仿真中启用或禁用对抗运动先验进行策略训练，然后部署到物理机器人上。

参考轨迹生成、奖励权重参数以及 AMP 配置。

1) *状态空间*：策略在每个时间步接收的观测空间 t 由本体感觉状态、参考信号和指令输入组成。每帧观测向量 $o_t \in \mathbb{R}^{50}$ 由以下分量拼接而成：

- **基础线速度** $v_b \in \mathbb{R}^3$ ：以机器人本体坐标系表示，缩放因子为 2.0。
- **速度指令** $c \in \mathbb{R}^3$ ： x 和 y 方向的目标线速度以及绕 z 轴的角速度。
- **相位编码** $\phi \in \mathbb{R}^2$ ：当前步态相位的正弦编码 $[\sin(2\pi\phi_t), \cos(2\pi\phi_t)]$ 。
- **基础角速度** $\omega_b \in \mathbb{R}^3$ ：以本体坐标系表示，缩放因子为 0.25。
- **投影重力向量** $g_p \in \mathbb{R}^3$ ：世界坐标系下的重力向量 $[0, 0, -1]^\top$ 经过基座方向旋转后的结果。
- **关节位置误差** $q_e \in \mathbb{R}^{12}$ ：当前关节位置与默认站立构型之间的差值。

- **Joint velocities** $\dot{q} \in \mathbb{R}^{12}$: scaled by 0.05.
- **Previous action** $a_{t-1} \in \mathbb{R}^{12}$: the action executed at the previous timestep.

The phase variable $\phi_t \in [0, 1)$ is advanced as $\phi_{t+1} = (\phi_t + \Delta t / T_{\text{cycle}}) \bmod 1$, where T_{cycle} is the gait-specific cycle time.

The reference joint trajectory is encoded through the phase variable. For periodic gaits (walking, goose-stepping, running, stair climbing), the reference is sinusoidal:

$$s = \sin(2\pi\phi_t) \quad (1)$$

$$q_{\text{ref}}^{\text{L,hip}} = -s \cdot \mathcal{K}_{s<0} \quad (2)$$

$$q_{\text{ref}}^{\text{L,knee}} = 2s \cdot \sigma \cdot \mathcal{K}_{s<0} \quad (3)$$

where σ is the target joint position scale (0.26 for walking and running, 0.28 for goose-stepping), and the right leg mirrors the left with opposite sign when $s > 0$. Each gait modulates this pattern through its $(T_{\text{cycle}}, \text{stance_ratio})$ pair: walking (0.8 s, 0.6), goose-stepping (0.7 s, 0.45), running (0.4 s, 0.35), stair climbing (0.7 s, 0.6).

For the jumping gait, a piecewise 4.0 s cycle trajectory is used instead, with five phases: squat ($\phi < 0.30$, $\sin^2$ knee flexion), takeoff (0.30–0.42, rapid extension), flight (0.42–0.48, hold extension), landing (0.48–0.75, impact absorption and recovery), and stand ($\phi \geq 0.75$, default pose). A curriculum progressively increases squat depth over 2000–10000 PPO iterations.

To capture temporal dynamics, the policy receives a history of $N = 21$ consecutive observation frames, stacked into a single input of dimension $50 \times 21 = 1050$. This enables implicit estimation of velocities and accelerations without explicit derivative computation.

The critic receives an extended observation s_t that additionally includes privileged simulation-only information: base linear and angular velocity, projected gravity, commands, joint position error, joint velocities, actions, phase encoding, friction coefficients, foot contact masks, stance masks, joint tracking error, push forces, foot vertical velocities, and foot clearance. This privileged vector (73 dimensions per frame) is stacked over 5 frames and concatenated with terrain height measurements (187 points), yielding 552 total dimensions.

2) *Action Space*: The policy outputs a 12-dimensional action vector $a_t \in \mathbb{R}^{12}$, corresponding to target joint position offsets for the 12 actuated DOFs (6 per leg: hip pitch, hip roll, hip yaw, knee, ankle pitch, ankle roll). The target joint position is computed as:

$$q_{\text{target}} = a_t \cdot k_{\text{action}} \cdot k_{\text{gain}} + q_{\text{default}} \quad (4)$$

where k_{action} is the action scale factor (0.25 for walking/stair climbing, 0.3 for running, 0.4 for goose-stepping) and $k_{\text{gain}} = \tau_{\text{max}} / K_P$ is a per-joint gain that normalizes the action output relative to the joint’s stiffness and torque limit.

The target positions are sent to a PD controller:

$$\tau = K_P(q_{\text{target}} - q) - K_D\dot{q} \quad (5)$$

The proportional gains K_P range from 75 to 280 N·m/rad for walking-type gaits (hip pitch and knee at 280, hip roll at 240, hip yaw at 75, ankle joints at 75), with derivative gains K_D from 5 to 14 N·m·s/rad. Goose-stepping employs significantly higher stiffness (hip pitch 450, knee 800, hip roll 300, hip yaw 150, ankle joints 200) and damping (K_D up to 40 on the knee), while running uses moderate increases (hip pitch and knee at 300, hip roll 240, hip yaw 100, ankle joints 80). Computed torques are clipped to per-joint maximum limits (60–150 N·m) specified in the URDF effort field. The policy executes at 50 Hz while the PD controller runs at 200 Hz (decimation factor of 4). An exponential moving average filter smooths actions, and a random action delay of 5–10 timesteps simulates real-world latency.

3) *Reward Function*: All five gaits share a unified reward formulation. The total reward at each timestep is a weighted sum:

$$r_{\text{total}} = \sum_i w_i r_i \quad (6)$$

The reward terms are organized into four categories:

(1) Joint tracking reward. The primary objective tracks the reference trajectory. For periodic gaits, the tracking reward for hip pitch and knee joints is computed as a dual-exponential sum:

$$r_{\text{track}} = 0.15 \sum_{j \in \{\text{hip,knee}\}} \sum_{i \in \{\text{L,R}\}} \left(e^{-4e_{ij}^2} + e^{-20e_{ij}^2} \right) \quad (7)$$

where $e_{ij} = q_{ij} - q_{\text{ref},ij}$ is the joint error, with the knee term masked to apply only during swing phase. Walking and running use a general tracking weight of $w_{\text{track}} = 2.0$, goose-stepping uses 4.0 (with tighter $\sigma = 0.15$ vs. 0.2), and a separate knee-only reward adds $w = 3.0$ (walking) or 25.0 (goose-stepping). For jumping, a single tracking reward ($w = 20.0$) computes mean squared error across all 12 joints.

(2) Base balance and posture reward. Includes orientation reward ($w = 3$ for walking and running, 5.0 for goose-stepping, 15.0 for jumping), base height reward ($w = 2$), and velocity tracking for linear ($w = 5.0$ walking, 8.0 goose-stepping, 20.0 running, 0.0 jumping) and angular ($w = 3.5$) velocity.

(3) Motion smoothness reward. Action smoothness penalizes first- and second-order changes ($w = -0.01$ for walking and running, -0.001 for goose-stepping), joint velocity ($w = -10^{-4}$), and rated torque ($w = -0.02$ per side for walking, -0.03 for goose-stepping, -0.05 for running). Goose-stepping adds leg straightness ($w = 12.0$), calf lift ($w = 8.0$), and foot kick ($w = 6.0$) terms.

(4) Safety and gait-specific rewards. Fall termination when the base contacts ground or $|\text{roll}| > 0.8$ rad, $|\text{pitch}| > 1.0$ rad. Knee collision penalties are applied. Periodic gaits include feet swing height ($w = 2$) and alternate foot swing ($w = 6.0$ for goose-stepping). Jumping uses specialized rewards: jump height targeting ($w = 150.0$, Gaussian reward centered at 0.275 m with a standing-phase penalty to prevent spurious small hops), vertical takeoff velocity ($w = 120.0$, piecewise nonlinear: linear ramp below 1.0 m/s, quadratic

- **关节速度** $\dot{q} \in \mathbb{R}^{12}$: 缩放因子为 0.05。
- **上一动作** $a_{t-1} \in \mathbb{R}^{12}$: 上一时间步执行的动作。

相位变量 $\phi_t \in [0, 1$ 按 $\phi_{t+1} = (\phi_t + \Delta t / T_{\text{cycle}}) \bmod 1$ 推进，其中 T_{cycle} 是步态特定的周期时间。

参考关节轨迹通过相位变量进行编码。对于周期性步态（行走、正步走、跑步、爬楼梯），参考轨迹为正弦形式：

$$s = \sin(2\pi\phi_t) \quad (1)$$

$$q_{\text{ref}}^{\text{L,hip}} = -s \cdot \mathcal{K}_{s<0} \quad (2)$$

$$q_{\text{ref}}^{\text{L,knee}} = 2s \cdot \sigma \cdot \mathcal{K}_{s<0} \quad (3)$$

其中 σ 是目标关节位置缩放因子（行走和跑步为 0.26, 正步走为 0.28），当 $s > 0$ 时右腿与左腿镜像对称且符号相反。每种步态通过其 $(T_{\text{cycle}}, \text{支撑相比例})$ 组合调节该模式：行走（0.8 秒, 0.6），正步走（0.7 秒, 0.45），跑步（0.4 秒, 0.35），爬楼梯（0.7 秒, 0.6）。

对于跳跃步态，采用分段式 4.0 秒周期轨迹，包含五个阶段：下蹲（ $\phi < 0.30$, 正弦² 膝关节屈曲）、起跳（0.30–0.42, 快速伸展）、腾空（0.42–0.48, 保持伸展）、落地（0.48–0.75, 冲击吸收与恢复）和站立（ $\phi \geq 0.75$, 默认姿态）。课程学习在 2000–10000 次 PPO 迭代中逐步增加下蹲深度。

为了捕捉时间动态，策略接收一个由 $N = 21$ 个连续观测帧组成的历史记录，堆叠成一个维度为 $50 \times 21 = 1050$ 的单一输入。这使得无需显式计算导数即可隐式估计速度和加速度。

评论家接收一个扩展观测 s_t ，其中额外包含仅模拟可用的特权信息：基座线速度和角速度、投影重力、指令、关节位置误差、关节速度、动作、相位编码、摩擦系数、足部接触掩码、站立掩码、关节跟踪误差、推力、足部垂直速度和足部离地间隙。该特权向量（每帧 73 维）在 5 帧上堆叠，并与地形高度测量值（187 个点）拼接，总共产生 552 维。

2) 动作空间：策略输出一个 12 维的动作向量 $a_t \in \mathbb{R}^{12}$ ，对应于 12 个驱动自由度（每条腿 6 个：髋关节俯仰、髋关节横滚、髋关节偏航、膝关节、踝关节俯仰、踝关节横滚）的目标关节位置偏移。目标关节位置计算如下：

$$q_{\text{target}} = a_t \cdot k_{\text{action}} \cdot k_{\text{gain}} + q_{\text{default}} \quad (4)$$

其中 k_{action} 是动作缩放因子（行走 / 爬楼梯为 0.25, 跑步为 0.3, 正步走为 0.4）， $k_{\text{gain}} = \tau_{\text{max}} / K_P$ 是每个关节的增益，用于根据关节的刚度和扭矩限制对动作输出进行归一化。

目标位置被发送到 PD 控制器：

$$\tau = K_P(q_{\text{target}} - q) - K_D\dot{q} \quad (5)$$

对于行走类步态，比例增益 K_P 范围为 75 至 280 N·m/rad（髋关节俯仰和膝关节为 280，髋关节横滚为 240，髋关节偏航为 75，踝关节为 75），微分增益 K_D 范围为 5 至 14 N·m·s/rad。正步走采用显著更高的刚度（髋关节俯仰 450, 膝关节 800, 髋关节横滚 300, 髋关节偏航 150, 踝关节 200）和阻尼（膝关节 K_D 高达 40），而跑步则采用适中的增加（髋关节俯仰和膝关节为 300, 髋关节横滚 240, 髋关节偏航 100, 踝关节 80）。计算出的扭矩被裁剪到 URDF 中 effort 字段指定的每个关节最大限制（60–150 N·m）。策略以 50 Hz 执行，而 PD 控制器以 200 Hz 运行（抽取因子为 4）。指数移动平均滤波器平滑动作，5–10 个时间步的随机动作延迟模拟真实世界的延迟。

3) 奖励函数：所有步态共享统一的奖励公式。每个时间步的总奖励为加权和：

$$r_{\text{total}} = \sum_i w_i r_i \quad (6)$$

奖励项分为四类：

(1) 关节跟踪奖励。主要目标跟踪参考轨迹。对于周期性步态，髋关节和膝关节的跟踪奖励计算为双指数和：

$$r_{\text{track}} = 0.15 \sum_{j \in \{\text{hip,knee}\}} \sum_{i \in \{\text{L,R}\}} \left(e^{-4e_{ij}^2} + e^{-20e_{ij}^2} \right) \quad (7)$$

其中 $e_{ij} = q_{ij} - q_{\text{ref},ij}$ 为关节误差，膝关节项仅在摆动阶段施加掩码。行走和跑步使用通用跟踪权重 $w_{\text{track}} = 2.0$ ，鹅步使用 4.0（ $\sigma = 0.15$ 与 0.2 相比更严格），单独的膝关节奖励增加 $w = 3.0$ （行走）或 25.0（鹅步）。对于跳跃，单一跟踪奖励（ $w = 20.0$ ）计算所有 12 个关节的均方误差。

(2) 基座平衡与姿态奖励。包括朝向奖励（行走和跑步为 $w = 3$ ，鹅步为 5.0，跳跃为 15.0）、基座高度奖励（ $w = 2$ ），以及线速度跟踪（行走 $w = 5.0$ 、鹅步 8.0、跑步 20.0、跳跃 0.0）和角速度跟踪（ $w = 3.5$ ）。

(3) 运动平滑性奖励。动作平滑性惩罚一阶和二阶变化（行走和跑步为 $w = -0.01$ ，鹅步为 -0.001 ）、关节速度（ $w = -10^{-4}$ ）以及额定扭矩（行走每侧 $w = -0.02$ ，鹅步 -0.03 ，跑步 -0.05 ）。鹅步额外增加腿部伸直度（ $w = 12.0$ ）、小腿抬起（ $w = 8.0$ ）和脚部踢出（ $w = 6.0$ ）项。

(4) 安全性与步态特定奖励。当基座接触地面或 |横滚角| > 0.8 弧度、|俯仰角| > 1.0 弧度时触发摔倒终止。施加膝盖碰撞惩罚。周期性步态包括足部摆动高度（ $w = 2$ ）和交替足部摆动（ $w = 6.0$ 用于正步走）。跳跃使用专门奖励：跳跃高度目标（ $w = 150.0$, 以 0.275 米为中心的高斯奖励，并带有站立阶段惩罚以防止意外小跳），垂直起飞速度（ $w = 120.0$, 分段非线性：低于 1.0 米 / 秒时线性斜坡，二次

above), and synchronized feet contact ($w = 20.0$), while penalizing horizontal velocity ($w = -15.0$).

The key design principle is that *all five gaits use the same reward formulation*—diversity arises from reference specification, weight tuning, and gait-specific terms rather than structural differences.

C. Selective AMP Strategy

1) *Brief Review of AMP*: Adversarial Motion Prior (AMP) combines adversarial imitation learning with reinforcement learning to produce motions that resemble a set of reference demonstrations. The method introduces a discriminator network D_ϕ trained to distinguish between state transitions from expert motion clips and state transitions produced by the policy during execution.

The discriminator architecture is a 3-layer MLP with hidden dimensions [1024, 512, 256] and ReLU activations. The input to the discriminator is the concatenation of the current state s_t and the next state s_{t+1} , capturing a single state transition. The discriminator outputs a scalar logit $D(s_t, s_{t+1})$, which is converted into a reward signal:

$$r_{\text{AMP}} = \alpha \cdot \max \left(0, 1 - \frac{1}{4} (D(s_t, s_{t+1}) - 1)^2 \right) \quad (8)$$

where α is a per-gait coefficient scaling the AMP reward magnitude. This reward is then linearly interpolated with the task reward:

$$r_{\text{combined}} = (1 - \beta) \cdot r_{\text{AMP}} + \beta \cdot r_{\text{task}} \quad (9)$$

where $\beta \in [0, 1]$ controls the balance between AMP regularization and task optimization.

The discriminator is trained with a binary cross-entropy loss combined with a gradient penalty for stability:

$$\mathcal{L}_D = \mathbb{E}_{\text{policy}}[(D+1)^2] + \mathbb{E}_{\text{expert}}[(D-1)^2] + \lambda \cdot \mathbb{E}_{\text{expert}}[\|\nabla_{(s,s')} D\|^2] \quad (10)$$

where the gradient penalty coefficient $\lambda = 10$. The reference motion data consists of captured human locomotion sequences, processed into 200,000 preloaded state transitions.

2) *Our Selection Criterion*: The central insight of our approach is that AMP’s regularization effect is beneficial for some gaits but detrimental for others. We propose a per-gait selection criterion based on motion characteristics:

AMP is applied to: Walking, goose-stepping, and stair climbing. These gaits are periodic with well-defined, repeating motion cycles and moderate joint amplitudes, where stability is the primary concern. For these gaits, the reference motion distribution closely matches the desired policy output, and the adversarial reward provides a stabilizing inductive bias that accelerates convergence, suppresses erratic behavior, and produces smoother joint trajectories compatible with real actuator dynamics. The AMP coefficient α and interpolation β are tuned per gait: walking and stair climbing use $\alpha = 0.3$, $\beta = 0.8$; goose-stepping uses $\alpha = 0.6$, $\beta = 0.7$

(higher α to enforce the distinctive straight-leg style; lower β to allow more AMP influence).

AMP is NOT applied to: Running and jumping. These gaits involve large joint excursions that may exceed the range of available reference data, and require rapid acceleration and explosive dynamics. For running, setting $\alpha = 0.0$ and $\beta = 1.0$ removes AMP entirely, giving the policy full freedom to discover high-speed strategies using the actuator’s complete torque capacity (up to 150 N·m on knee joints). Similarly, jumping uses $\alpha = 0.0$, $\beta = 1.0$, and disables the symmetry loss (sym_loss = False) to allow asymmetric force generation during takeoff. In both cases, the AMP discriminator would penalize the policy for deviating from the reference distribution, actively discouraging the motion extremes needed for sufficient speed or jump height.

In summary: *AMP improves stability but limits expressiveness; we select its use adaptively based on gait characteristics*. Table I summarizes the configuration for each gait.

TABLE I
SELECTIVE AMP CONFIGURATION PER GAIT.

Gait	α	β	Sym Loss
Walking	0.3	0.8	True
Goose-stepping	0.6	0.7	True
Running	0.0	1.0	True
Stair climbing	0.3	0.8	True
Jumping	0.0	1.0	False

D. Training Pipeline

1) *Simulation Training*: All policies are trained using the Proximal Policy Optimization (PPO) algorithm [9] in the Isaac Gym physics simulator [5] with 4096 parallel environments. The actor network uses an MLP with hidden layers [512, 256, 128] and ELU activation. The critic network shares the same architecture. The PPO algorithm runs 5 learning epochs over 4 mini-batches per iteration, with a learning rate of 5×10^{-4} , discount factor $\gamma = 0.99$, and GAE parameter $\lambda = 0.95$. The KL divergence target is set to 0.01 with an adaptive learning rate schedule. Training proceeds for up to 40,000 iterations, with the empirical observation normalizer updated online during training [10].

2) *Domain Randomization*: To bridge the simulation-to-reality gap, we apply extensive domain randomization during training. The following parameters are randomized at the environment level each episode:

- **Contact properties**: Static and dynamic friction uniformly sampled from [0.2, 1.5]; restitution from [0.0, 1.0].
- **Base mass**: Perturbed by $[-2, +5]$ kg from the nominal mass.
- **Link inertia**: Scaled by [0.5, 1.8] independently per link.
- **Center of mass**: Displaced by $[-0.1, 0.1]$ m in each axis.
- **Motor strength**: Randomized to [0.7, 1.0] (walking/stair climbing), [0.8, 1.2] (running/goose-stepping), or [0.9, 1.1] (jumping).

上述), 以及同步的足部接触 ($w = 20.0$), 同时惩罚水平速度 ($w = -15.0$).

关键设计原则是, 所有步态使用相同的奖励公式——多样性源于参考规范、权重调整和步态特定项, 而非结构差异。

C. 选择性对抗运动先验策略

1) AMP 简要回顾: 对抗运动先验 (AMP) 将对抗模仿学习与强化学习相结合, 以生成类似于一组参考演示的动作。该方法引入了一个判别器网络 D_ϕ , 该网络经过训练, 能够区分来自专家运动片段的状态转换与策略在执行过程中产生的状态转换。

判别器架构是一个具有隐藏维度 [1024, 512, 256] 和 ReLU 激活函数的三层 MLP。判别器的输入是当前状态 s_t 与下一状态 s_{t+1} 的拼接, 捕捉单个状态转换。判别器输出一个标量 $\text{logit}D(s_t, s_{t+1})$, 该值被转换为奖励信号:

$$r_{\text{AMP}} = \alpha \cdot \max \left(0, 1 - \frac{1}{4} (D(s_t, s_{t+1}) - 1)^2 \right) \quad (8)$$

其中 α 是缩放 AMP 奖励幅度的每个步态的系数。

该奖励随后与任务奖励进行线性插值:

$$r_{\text{combined}} = (1 - \beta) \cdot r_{\text{AMP}} + \beta \cdot r_{\text{task}} \quad (9)$$

其中 $\beta \in [0, 1]$ 控制 AMP 正则化与任务优化之间的平衡。

判别器通过二元交叉熵损失结合梯度惩罚进行训练, 以保证稳定性:

$$\mathcal{L}_D = \mathbb{E}_{\text{policy}}[(D+1)^2] + \mathbb{E}_{\text{expert}}[(D-1)^2] + \lambda \cdot \mathbb{E}_{\text{expert}}[\|\nabla_{(s,s')} D\|^2] \quad (10)$$

其中梯度惩罚系数 $\lambda = 10$ 。参考运动数据由采集到的人体运动序列组成, 经过处理后得到 200,000 个预加载的状态转移样本。

2) 我们的选择标准: 我们方法的核心洞察在于, AMP 的正则化效果对某些步态有益, 但对其他步态有害。我们提出了一种基于运动特征的逐步态选择标准:

AMP 应用于: 行走、正步走和爬楼梯。这些步态具有周期性, 运动周期明确且重复, 关节幅度适中, 稳定性是首要关注点。对于这些步态, 参考运动分布与期望的策略输出高度匹配, 对抗性奖励提供了稳定的归纳偏置, 加速收敛、抑制异常行为, 并生成与真实执行器动力学兼容的更平滑的关节轨迹。AMP 系数 α 和插值系数 β 针对每种步态进行调整: 行走和爬楼梯使用 $\alpha = 0.3$ 、 $\beta = 0.8$; 正步走使用 $\alpha = 0.6$ 、 $\beta = 0.7$ 。

(更高的 α 以强化独特的直腿风格; 更低的 β 以允许更大的 AMP 影响)。

AMP 不适用于: 奔跑和跳跃。这些步态涉及较大的关节运动范围, 可能超出可用参考数据的范围, 并且需要快速加速和爆发性动力学。对于奔跑, 设置 $\alpha = 0.0$ 和 $\beta = 1.0$ 完全移除 AMP, 使策略能够自由利用执行器的全部扭矩能力 (膝关节扭矩高达 150 N·m) 来发现高速策略。类似地, 跳跃使用 $\alpha = 0.0$ 、 $\beta = 1.0$, 并禁用对称损失 (sym_loss = False), 以允许在起跳过程中产生不对称的力。在这两种情况下, AMP 判别器会因偏离参考分布而惩罚策略, 从而主动抑制达到足够速度或跳跃高度所需的极端运动。

总结: AMP 提高了稳定性但限制了表现力; 我们根据步态特征自适应地选择其使用。表 I 总结了每种步态的配置。

表 I 每种步态的选择性 AMP 配置。

Gait	α	β	Sym Loss
Walking	0.3	0.8	True
Goose-stepping	0.6	0.7	True
Running	0.0	1.0	True
Stair climbing	0.3	0.8	True
Jumping	0.0	1.0	False

D. 训练流程

1) 仿真训练: 所有策略均使用近端策略优化 (PPO) 算法 [9] 在 Isaac Gym 物理仿真器 [5] 中训练, 配备 4096 个并行环境。演员网络使用具有隐藏层 [512, 256, 128] 和 ELU 激活函数的多层感知机。评论家网络共享相同架构。PPO 算法每次迭代运行 5 个学习周期, 分为 4 个小批量, 学习率为 5×10^{-4} , 折扣因子为 $\gamma = 0.99$, GAE 参数为 $\lambda = 0.95$ 。KL 散度目标设为 0.01, 并采用自适应学习率调度。训练最多进行 40,000 次迭代, 经验观测归一化器在训练过程中在线更新 [10]。

2) 域随机化: 为了弥合仿真与现实的差距, 我们在训练过程中应用了广泛的域随机化。以下参数在每个回合的环境层面进行随机化:

• **接触属性**: 静摩擦和动摩擦统一从 [0.2, 1.5] 中采样; 回弹系数从 [0.0, 1.0] 中采样。

• **基座质量**: 在标称质量基础上扰动 $[-2, +5]$ 千克。

• **连杆惯性**: 每个连杆独立按 [0.5, 1.8] 缩放。

• **质心**: 在每个轴上偏移 $[-0.1, 0.1]$ 米。

• **电机强度**: 随机化为 [0.7, 1.0] (行走 / 爬楼梯)、[0.8, 1.2] (跑步 / 正步走) 或 [0.9, 1.1] (跳跃)。

- **PD gains:** K_P scaled by [0.8, 1.2], K_D scaled by [0.8, 1.2].
- **Joint properties:** Joint friction [0.3, 1.5], damping [0.3, 4.0], armature [0.8, 1.2].
- **Action lag:** Actions delayed by a random number of timesteps from [5, 10], simulating control latency.
- **External disturbances:** Random impulses applied to the base with maximum linear velocity 2.0 m/s and angular velocity 1.5 rad/s, at intervals of 10 seconds.
- **Initial state:** Joint positions scaled by [0.5, 1.5] and offset by $[-0.1, 0.1]$ rad at reset.

A curriculum is employed for action lag: the minimum delay is gradually increased from 2 to 5 timesteps over the course of training, starting from a milder setting to facilitate initial learning and progressively increasing the difficulty.

3) *Sim-to-Real Transfer*: Trained policies are deployed on the physical humanoid robot using zero-shot transfer—no fine-tuning on real hardware is performed. The deployment pipeline consists of:

- 1) **Policy export:** The trained actor network is exported to ONNX format with the empirical observation normalizer as a single inference module, and loaded via ONNX Runtime on the robot’s onboard computer.
- 2) **Onboard execution:** The policy runs onboard the robot’s computer at 50 Hz, receiving proprioceptive observations from an external IMU (angular velocity and orientation) and joint encoders (position and velocity). Since base linear velocity is not directly measurable on the hardware, the corresponding observation channels are set to zero, relying on the observation history stack to implicitly capture velocity information.
- 3) **PD control:** Joint torques are computed by the onboard PD controller at 500 Hz using the target positions output by the policy. A first-order low-pass filter with cutoff frequency 50 Hz smooths the policy actions before conversion to joint targets.

The domain randomization applied during training ensures that the policy is robust to the discrepancies between simulated and real dynamics, including differences in motor response, contact behavior, and sensor noise.

III. EXPERIMENTS

The video of the experiments can be found at <https://linqi-ye.github.io/video/lite.mp4>

A. Experimental Setup

Experiments were conducted on the Lite_11 humanoid lower-body platform, which provides 12 actuated joints spanning the bilateral hip, knee, and ankle degrees of freedom. All policies were trained in Isaac Gym and then deployed on the physical robot for validation. We considered five locomotion modes: walking, goose-stepping, stair climbing, running, and jumping. Stair climbing includes both ascent and descent, whereas jumping consists of in-place two-foot takeoff followed by landing recovery. All tasks share the same policy interface: a 12-dimensional action vector, a 50-dimensional per-frame observation, and a 21-frame history

stack, resulting in a 1050-dimensional policy input. Consequently, gait diversity is induced by reference trajectories, reward allocation, and motion-prior configuration rather than by changing the control architecture.

All tasks use a unified simulation and control backbone. The physics step is 0.005 s, and the policy is updated every four simulation steps, yielding an effective control period of 0.02 s. Low-level execution is handled by a PD position controller, which converts the policy-produced joint target offsets into actuation commands. PPO is used for all experiments. Except for jumping, which uses a slightly higher entropy coefficient to accommodate stronger exploration demands, the same base hyperparameters are shared across tasks. The main training settings are summarized in Table II.

Task environments and prior settings are defined according to gait characteristics. Walking, goose-stepping, and running are trained primarily on flat or low-complexity terrain, stair climbing is trained on triangle-mesh stairs with curriculum learning, and jumping is trained on flat terrain to emphasize vertical impulse generation, aerial stabilization, and landing recovery. Selective AMP is central to the experimental design: walking, goose-stepping, and stair climbing belong to the AMP group, whereas running and jumping are trained without AMP. This partition reflects the observation that periodic, stability-oriented motions benefit from motion priors, whereas highly dynamic motions depend more strongly on expressive freedom and task-driven optimization. To improve sim-to-real transfer robustness, domain randomization is applied throughout training, including friction, added mass, center-of-mass perturbation, motor strength, PD gains, inertia, joint friction, joint damping, initial joint scaling and bias, and action delay. Stair climbing uses a stronger terrain curriculum, while jumping adopts a more conservative randomization range.

B. Evaluation Metrics

To evaluate multi-gait policies from both optimization and execution perspectives, we consider joint tracking quality, convergence speed, task reliability, training stability, and posture stability. Performance is not judged solely by whether a motion emerges, but by whether the learned motion adheres to the intended structure, converges stably during training, and remains controllable during execution.

The joint tracking metric characterizes how well the policy preserves the intended kinematic organization of each gait. For walking, goose-stepping, and stair climbing, it primarily reflects coordination between stance and swing phases; for running and jumping, it reflects the ability to maintain the target motion pattern under high-speed or explosive dynamics. Convergence steps quantify the optimization cost required to reach a stable regime. Tasks that stabilize within fewer iterations are easier to optimize under the unified formulation, whereas tasks that require more iterations place stronger demands on contact switching, posture recovery, or temporal coordination.

At execution time, success rate and fall rate describe task reliability. A high success rate indicates that the robot can

- **PD 增益** : K_P 缩放 [0.8, 1.2], K_D 缩放 [0.8, 1.2]。

- **关节属性** : 关节摩擦 [0.3, 1.5], 阻尼 [0.3, 4.0], 电枢 [0.8, 1.2]。

- **动作延迟** : 动作延迟随机时间步数, 范围从 [5, 10], , 模拟控制延迟。
- **外部扰动** : 对基座施加随机冲量, 最大线速度 2.0 米 / 秒, 角速度 1.5 弧度 / 秒, 间隔 10 秒。

- **初始状态** : 关节位置在重置时缩放 [0.5, 1.5] 并偏移 $[-0.1, 0.1]$ 弧度。

对动作延迟采用课程学习：在训练过程中, 最小延迟从 2 个时间步逐渐增加到 5 个时间步, 从较温和的设置开始以促进初始学习, 并逐步增加难度。

3) 仿真到现实迁移：训练好的策略通过零样本迁移部署在物理人形机器人上 —— 无需在真实硬件上进行微调。部署流程包括：

- 1) 策略导出：将训练好的演员网络与经验观测归一化器一起导出为 ONNX 格式, 作为一个单一推理模块, 并通过机器人板载计算机上的 ONNX 运行时加载。
- 2) 板载执行：策略以 50 Hz 的频率在机器人板载计算机上运行, 接收来自外部 IMU （角速度和姿态）和关节编码器（位置和速度）的本体感受观测。由于硬件上无法直接测量基座线速度, 相应的观测通道设置为零, 依靠观测历史堆栈隐式捕获速度信息。

- 3) PD 控制：关节扭矩由板载 PD 控制器以 500 Hz 的频率计算, 使用策略输出的目标位置。在转换为关节目标之前, 采用截止频率为 50 Hz 的一阶低通滤波器对策略动作进行平滑处理。

训练过程中应用的域随机化确保了策略对仿真与现实动力学之间的差异具有鲁棒性, 包括电机响应、接触行为和传感器噪声等方面的差异。

III. 实验

实验视频可在 <https://linqi-ye.github.io/video/lite.mp4> 查看

A. 实验设置

实验在 Lite 11 仿人下肢平台上进行, 该平台提供 12 个驱动关节, 涵盖双侧髋、膝、踝关节自由度。所有策略均在 Isaac Gym 中训练, 随后部署至实体机器人进行验证。我们考虑了五种运动模式：行走、正步、爬楼梯、奔跑和跳跃。爬楼梯包括上行与下行, 而跳跃则包含原地双脚起跳及落地恢复。所有任务共享相同的策略接口：12 维动作向量、 50 维单帧观测值及 21 帧历史数据。

堆叠, 产生一个 1050 维的策略输入。因此, 步态多样性是由参考轨迹、奖励分配和运动先验配置所诱导, 而非通过改变控制架构实现。

所有任务均采用统一的仿真与控制框架。物理步长为 0.005 秒, 策略每四个仿真步长更新一次, 有效控制周期为 0.02 秒。底层执行由 PD 位置控制器完成, 该控制器将策略生成的关节目标偏移量转换为驱动指令。所有实验均使用 PPO 算法。除跳跃任务采用略高的熵系数以适应更强的探索需求外, 其余任务共享相同的基础超参数。主要训练设置汇总于表 II。

任务环境与先验设置根据步态特征进行定义。行走、正步和跑步主要在平坦或低复杂度地形上训练, 爬楼梯采用三角形网格楼梯并结合课程学习进行训练, 跳跃则在平坦地形上训练, 以强调垂直冲量生成、空中稳定和落地恢复。选择性 AMP 是实验设计的核心：行走、正步和爬楼梯属于 AMP 组, 而跑步和跳跃则在不使用 AMP 的情况下训练。这种划分反映了以下观察：周期性、稳定性导向的运动受益于运动先验, 而高度动态的运动则更依赖于表达自由度和任务驱动优化。为提升仿真到现实的迁移鲁棒性, 训练全程应用域随机化, 包括摩擦力、附加质量、质心扰动、电机强度、PD 增益、惯性、关节摩擦、关节阻尼、初始关节缩放与偏置以及动作延迟。爬楼梯采用更强的地形课程, 而跳跃则采用更保守的随机化范围。

B. 评估指标

为了从优化和执行两个角度评估多步态策略, 我们考虑了关节跟踪质量、收敛速度、任务可靠性、训练稳定性和姿态稳定性。性能不仅仅取决于是否产生了运动, 还取决于学习到的运动是否符合预期的结构、在训练过程中是否稳定收敛, 以及在执行过程中是否保持可控。

关节跟踪指标表征了策略在多大程度上保留了每种步态预期的运动学组织。对于行走、正步和爬楼梯, 它主要反映支撑相和摆动相之间的协调性; 对于奔跑和跳跃, 它反映了在高速或爆发性动力学下维持目标运动模式的能力。收敛步数量化了达到稳定状态所需的优化成本。在统一公式下, 能在较少迭代次数内稳定的任务更容易优化, 而需要更多迭代次数的任务则对接触切换、姿态恢复或时间协调提出了更高的要求。

在执行时, 成功率和跌倒率描述了任务可靠性。高成功率表明机器人能够

TABLE II
MAIN TRAINING CONFIGURATIONS FOR THE FIVE GAIT TASKS.

Gait	# Envs	Obs. Dim.	Act. Dim.	Sim Step	Control Period	num.steps.per.env	Max Iterations	AMP Group
Walking	4096	1050	12	0.005 s	0.02 s	24	40001	AMP
Goose-stepping	4096	1050	12	0.005 s	0.02 s	24	20001	AMP
Stair climbing	4096	1050	12	0.005 s	0.02 s	24	40001	AMP
Running	2048	1050	12	0.005 s	0.02 s	24	40001	No AMP
Jumping	2048	1050	12	0.005 s	0.02 s	24	11000	No AMP

repeatedly sustain the desired gait, whereas a high fall rate typically indicates sensitivity to initial conditions, rhythm control, or dynamic recovery. Training error characterizes the smoothness of optimization: lower error generally corresponds to more stable parameter updates, whereas persistently large error suggests stronger oscillation and higher uncertainty. Posture stability measures whole-body control quality. Higher posture stability corresponds to smaller torso oscillation, smoother roll-pitch regulation, and more consistent phase transitions. These metrics are used below to compare both overall gait performance and the distinction between AMP and non-AMP training.

C. Results and Analysis

1) *Overall Performance of Five Gaits*: Under a unified policy structure, shared observation design, and common reward organization, the proposed framework learns and deploys five locomotion modes: walking, goose-stepping, stair climbing, running, and jumping. These tasks span the spectrum from low-dynamic periodic locomotion to highly dynamic explosive motion, thereby testing the scalability of the unified RL formulation across substantially different gait classes.

Qualitatively, walking yields a regular forward stepping pattern; goose-stepping exhibits the intended high leg lift and straight-knee swing; stair climbing maintains sufficient foot clearance and body posture during both ascent and descent; running produces shorter stance durations and stronger forward propulsion; and jumping realizes the full motion chain from squat preparation and rapid takeoff to aerial stabilization and landing recovery. Figure 2 presents representative real-robot image sequences for the five gaits. These results indicate that the unified framework is able not only to maintain balance, but also to generate motion patterns with clearly differentiated dynamic characteristics and stylistic features.

Table III summarizes the quantitative performance of the five gaits. The reported indicators correspond to joint tracking quality, convergence speed, task reliability, training stability, and posture stability. Because the reward composition and task difficulty differ across gaits, these values are intended to characterize training and execution behavior under the unified framework rather than to serve as strictly normalized physical quantities across tasks.

2) *Effect of AMP on Different Gaits*: This subsection examines the operating boundary of AMP across gait classes. In general, AMP is most effective for gaits with clear periodic structure and stability-first objectives, whereas it can

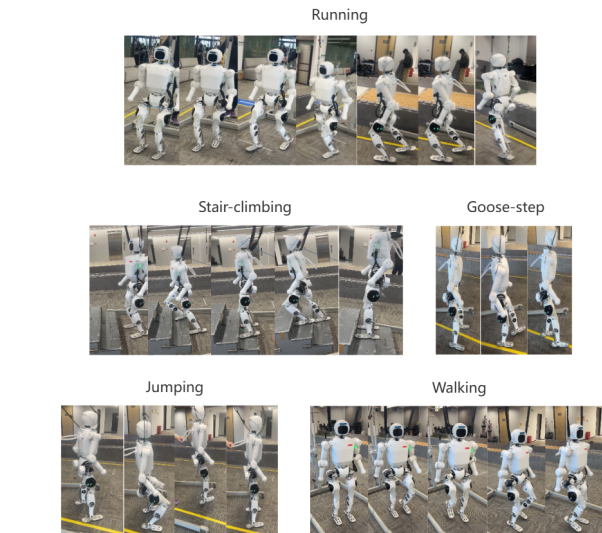


Fig. 2. Representative real-robot image sequences for the five learned gaits: walking, stair climbing, goose-stepping, jumping, and running.

become restrictive for highly dynamic and strongly explosive motions. Walking, goose-stepping, and stair climbing fall into the former category: they require stable stance–swing transitions, consistent joint coordination, and low postural fluctuation. For such tasks, motion priors reduce ineffective exploration and typically yield lower tracking error, smoother reward growth, and faster convergence.

Figure 3 compares goose-stepping with AMP and jumping without AMP as representative cases. Under AMP, goose-stepping exhibits a rapid increase in total reward, low training error, and a prompt decay in failure rate. The main contribution of AMP in this setting is not merely a higher terminal return, but the reinforcement of periodic structure and style consistency, which is especially important for a gait with explicit kinematic features such as high leg lift and straight-knee swing. For running, a strong AMP prior would not necessarily render the policy incapable of locomotion, but it would likely increase behavioral conservatism and bias the solution toward a more strongly regularized motion pattern. In that regime, the learned behavior would be more likely to resemble a smooth fast walk or a weak running pattern, rather than a genuinely explosive gait with clear aerial phases, strong forward propulsion, and large leg excursion. This observation suggests that, for highly dynamic locomotion, an excessively strong motion prior may improve motion regularity while simultaneously compressing the policy’s access to high-amplitude dynamic solutions. By contrast, jumping

表 II 五种步态任务的主要训练配置

Gait	# Envs	Obs. Dim.	Act. Dim.	Sim Step	Control Period	num.steps.per.env	Max Iterations	AMP Group
Walking	4096	1050	12	0.005 s	0.02 s	24	40001	AMP
Goose-stepping	4096	1050	12	0.005 s	0.02 s	24	20001	AMP
Stair climbing	4096	1050	12	0.005 s	0.02 s	24	40001	AMP
Running	2048	1050	12	0.005 s	0.02 s	24	40001	No AMP
Jumping	2048	1050	12	0.005 s	0.02 s	24	11000	No AMP

持续维持目标步态，而高跌倒率通常表明对初始条件、节奏控制或动态恢复的敏感性。训练误差表征优化的平滑性：较低的误差通常对应更稳定的参数更新，而持续较大的误差则表明更强的振荡和更高的不确定性。姿态稳定性衡量全身控制质量。较高的姿态稳定性对应较小的躯干振荡、更平滑的横滚 - 俯仰调节以及更一致的相位转换。以下使用这些指标来比较整体步态性能以及 AMP 与非 AMP 训练之间的差异。

C. 结果与分析

1) 五种步态的整体性能：在统一的策略结构、共享的观测设计和通用的奖励组织下，所提出的框架学习并部署了五种运动模式：行走、正步、爬楼梯、奔跑和跳跃。这些任务涵盖了从低动态周期性运动到高动态爆发性运动的范围，从而测试了统一强化学习公式在显著不同步态类别上的可扩展性。

定性来看，行走呈现出规则的向前迈步模式；正步表现出预期的高抬腿和直膝摆动；爬楼梯在上下楼梯过程中保持足够的足部间隙和身体姿态；奔跑产生更短的支撑相和更强的向前推进力；跳跃则实现了从下蹲准备、快速起跳到空中稳定和落地恢复的完整运动链。图 2 展示了五种步态的代表性真实机器人图像序列。这些结果表明，统一框架不仅能够维持平衡，还能生成具有清晰差异化动态特征和风格特征的运动模式。

表 III 总结了五种步态的量化性能。报告的指标对应关节跟踪质量、收敛速度、任务可靠性、训练稳定性和姿态稳定性。由于不同步态的奖励构成和任务难度存在差异，这些值旨在表征统一框架下的训练和执行行为，而非作为跨任务的严格归一化物理量。

2) AMP 对不同步态的影响：本小节考察 AMP 在不同步态类别上的运行边界。总体而言，AMP 对于具有清晰周期结构和以稳定性为首要目标的步态最为有效，而对于

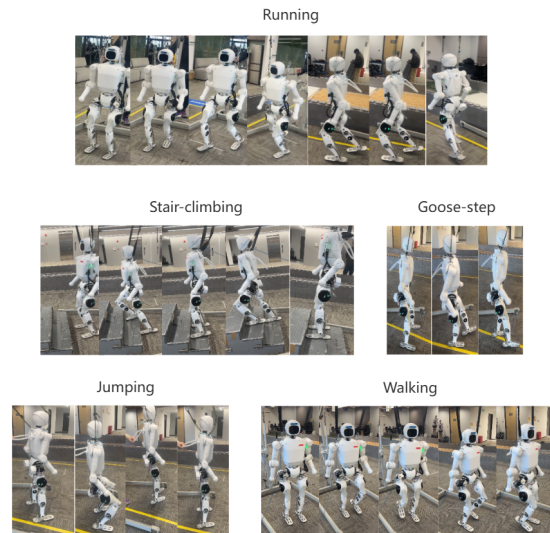


图 2. 五种学习步态的代表性真实机器人图像序列：行走、爬楼梯、正步走、跳跃和跑步。

高度动态和强爆发性运动则可能产生限制。行走、正步走和爬楼梯属于前一类：它们需要稳定的支撑 - 摆动转换、一致的关节协调以及低姿态波动。对于此类任务，运动先验减少了无效探索，通常能带来更低的跟踪误差、更平滑的奖励增长和更快的收敛。

图 3 以带 AMP 的正步走和不带 AMP 的跳跃作为代表性案例进行比较。在 AMP 作用下，正步走表现出总奖励的快速增加、训练误差低以及失败率的迅速衰减。AMP 在此场景中的主要贡献并非仅仅是更高的最终回报，而是强化了周期结构和风格一致性，这对于具有明确运动学特征（如高抬腿和直膝摆动）的步态尤为重要。对于跑步而言，强 AMP 先验不一定会使策略无法完成运动，但很可能会增加行为保守性，并使解决方案偏向于更强烈正则化的运动模式。在这种情况下，学习到的行为更可能类似于平滑的快走或弱跑步模式，而非具有清晰腾空阶段、强向前推进力和大幅腿部摆动的真正爆发性步态。这一观察表明，对于高度动态的运动，过强的运动先验可能在改善运动规律性的同时，压缩了策略获取高幅度动态解的空间。相比之下，跳跃

TABLE III QUANTITATIVE SUMMARY OF THE FIVE GAIT TASKS.						
Gait	Joint-Tracking Metric	Convergence Steps	Estimated Success Rate	Training Error	Posture-Stability Metric	Motion Characteristic
Walking	0.40	2500	95%	10.0	2.3	Periodic and stable forward locomotion
Goose-stepping	0.94	2500	97%	5.4	1.2	High leg lift and straight-knee swing
Stair climbing	1.00	5200	85%	1.0	0.95	Stair ascent/descent with recovery
Running	0.50	3000	96%	4.8	1.3	Short stance and strong propulsion
Jumping	15.5	3000	83%	6.0	8.6	Two-foot takeoff and landing recovery

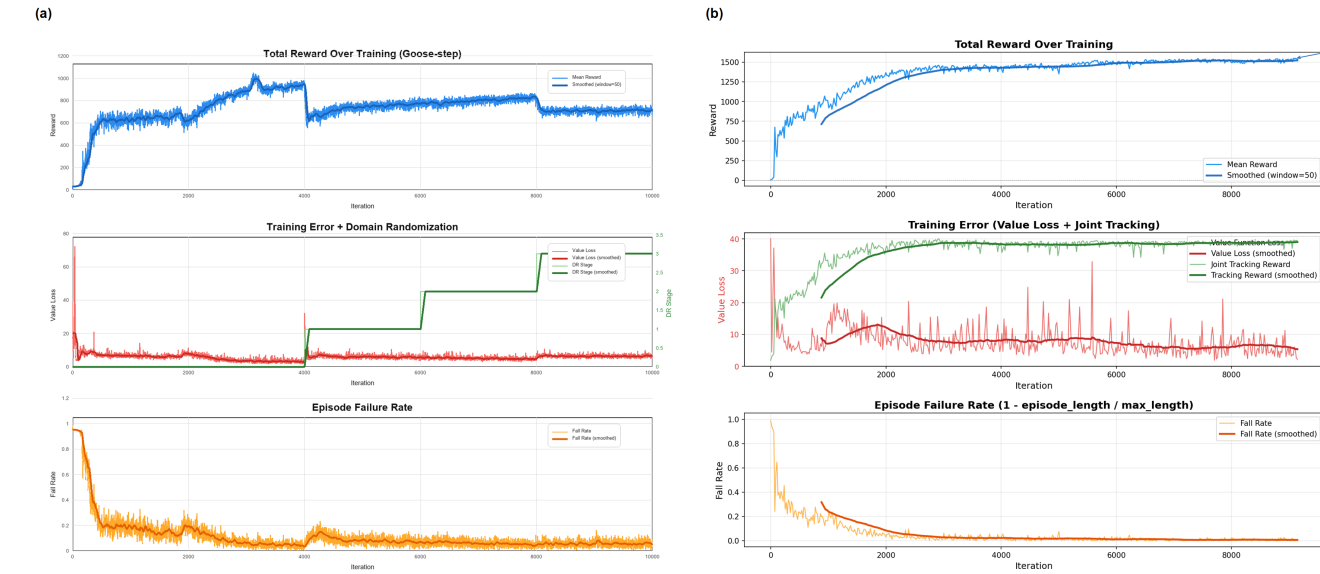


Fig. 3. Training curves used in the AMP-versus-no-AMP comparison. (a) Goose-stepping with AMP. (b) Jumping without AMP. Each panel reports total reward, training error or auxiliary metric, and failure rate.

depends on large motion amplitude, vertical impulse, and aerial freedom. Under the non-AMP setting, the task still attains sustained reward growth and low failure rate while the joint-tracking signal improves steadily. This indicates that task rewards alone are sufficient to drive effective takeoff, flight, and landing recovery when expressive exploration is preserved. These results support the view that AMP should be treated as a gait-dependent mechanism rather than a uniformly active module.

3) *Real-Robot Robustness and Practical Performance:* Beyond single demonstrations, the engineering value of the method is reflected in sustained execution on the physical robot. We therefore assess real-robot robustness from three perspectives: long-horizon continuity, consistency across repeated trials, and recovery under mild external disturbances or start-stop transitions.

The robustness criteria vary across gaits. For walking and goose-stepping, the emphasis lies in rhythm consistency and controlled torso oscillation; for stair climbing, in foot clearance, step placement, and posture recovery after stair transitions; for running, in stability during rapid short-stance switching; and for jumping, in synchronized takeoff, aerial posture control, landing absorption, and return to the next motion cycle. In particular, the jumping reference is explicitly divided into squat, takeoff, flight, landing, and standing-recovery phases, so robustness depends not only on leaving the ground, but also on the quality of aerial motion and

landing recovery.

Taken together, the real-robot deployments show that the learned policies are not limited to visually plausible motions in simulation, but exhibit the continuity and repeatability required for physical execution. This supports the practical viability of the proposed unified framework for multi-gait humanoid locomotion.

IV. CONCLUSION

This paper presented a unified reinforcement learning framework for multi-gait humanoid locomotion together with a selective AMP strategy. Within a common policy architecture and reward formulation, the proposed method learned five representative gaits, namely walking, goose-stepping, stair climbing, running, and jumping, and transferred them to a physical humanoid platform. The results indicate that motion priors are beneficial for periodic, stability-oriented gaits, but should be weakened or removed for highly dynamic behaviors that depend on expressive motion generation. More importantly, the proposed framework provides a practical and reproducible baseline for multi-gait learning on real humanoid hardware by combining a unified RL formulation, gait-dependent prior selection, and real-robot validation. Overall, this study suggests that practical humanoid mobility will depend not only on stronger policies, but also on principled methods for organizing diverse behaviors within a shared learning framework.

表 III 五种步态任务的定量总结。						
Gait	Joint-Tracking Metric	Convergence Steps	Estimated Success Rate	Training Error	Posture-Stability Metric	Motion Characteristic
Walking	0.40	2500	95%	10.0	2.3	Periodic and stable forward locomotion
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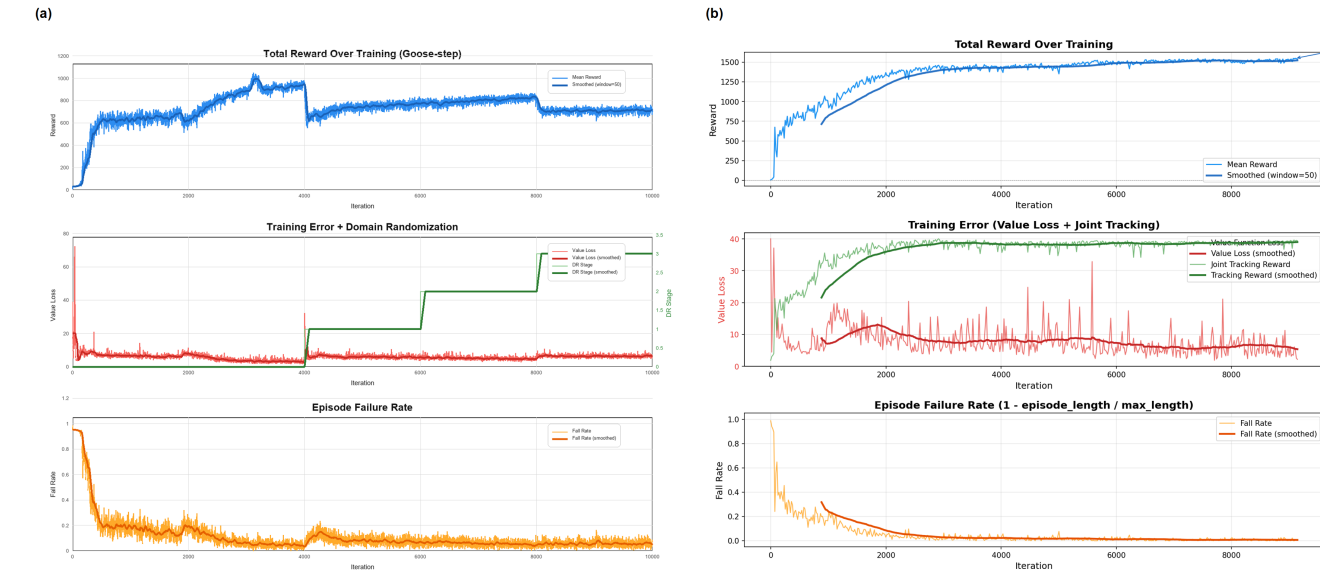


图 3. AMP 与无 AMP 对比中使用的训练曲线。(a) 带 AMP 的正步走。(b) 不带 AMP 的跳跃。每个面板报告总奖励、训练误差或辅助指标以及失败率。

依赖于较大的运动幅度、垂直冲量和空中自由度。在无 AMP 设置下，该任务仍能获得持续的奖励增长和低失败率，同时关节跟踪信号稳步改善。这表明，当保留富有表现力的探索时，仅凭任务奖励就足以驱动有效的起飞、飞行和着陆恢复。这些结果支持了 AMP 应被视为一种依赖于步态的机制，而非统一激活的模块的观点。

3) 真实机器人鲁棒性与实际性能：除了单次演示之外，该方法的工程价值体现在物理机器人上的持续执行。因此，我们从三个角度评估真实机器人的鲁棒性：长时域连续性、重复试验间的一致性，以及在轻微外部干扰或启停转换下的恢复能力。

不同步态的鲁棒性标准各不相同。对于行走和正步走，重点在于节奏一致性和受控的躯干摆动；对于爬楼梯，在于足部离地间隙、落脚点以及楼梯过渡后的姿态恢复；对于跑步，在于快速短支撑切换过程中的稳定性；对于跳跃，在于同步起跳、空中姿态控制、落地缓冲以及返回下一个运动周期。特别是，跳跃参考被明确划分为下蹲、起跳、飞行、落地和站立恢复阶段，因此鲁棒性不仅取决于离地，还取决于空中运动的质量和

落地恢复。

综合来看，真实机器人部署实验表明，学习到的策略不仅限于仿真中视觉上合理的运动，还展现出物理执行所需的连续性和可重复性。这验证了所提出的多步态人形机器人统一框架的实际可行性。

IV. 结论

本文提出了一种用于多步态人形机器人的统一强化学习框架，并结合了选择性对抗运动先验策略。在共同的策略架构和奖励函数设计下，所提出的方法学习了五种代表性步态——行走、正步、爬楼梯、奔跑和跳跃，并将其迁移至物理人形机器人平台。结果表明，运动先验对周期性、稳定性导向的步态有益，但对于依赖表现性运动生成的高动态行为，应削弱或移除运动先验。更重要的是，该框架通过结合统一的强化学习公式、步态依赖的先验选择以及真实机器人验证，为在真实人形硬件上实现多步态学习提供了实用且可复现的基线。总体而言，本研究表明，实用的人形机器人移动能力不仅依赖于更强的策略，还依赖于在共享学习框架内组织多样化行为的系统性方法。

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